1. A 67-year-old woman with non-Hodgkin lymphoma is being prepared for discharge from the hospital after treatment of a hypertensive crisis. She is currently receiving her first course of chemotherapy. One year ago, she received pneumococcal and influenza vaccines. Her blood pressure has responded well to lisinopril. Currently, her temperature is 37°C (98.6°F), pulse is 80/min, respirations are 18/min, and blood pressure is 140/86 mm Hg. Examination shows an enlarged left supraclavicular node. The spleen tip is palpated 5 cm below the left costal margin. Administration of which of the following vaccines is most appropriate for this patient?

A) Influenza
B) Measles-mumps-rubella
C) Meningococcal
D) Pneumococcal
E) Varicella

2. A 27-year-old woman comes to the physician because of vaginal discharge and itching for 1 week. She has been sexually active with multiple partners for 3 years, and they use condoms occasionally. Over the past 3 years, she has had three episodes of candidal vaginitis and three episodes of genital herpes. Pelvic examination shows a white vaginal discharge with a cottage-cheese consistency. Examination shows no other abnormalities. A Pap smear is obtained. Topical antifungal medications are prescribed. Which of the following is the most appropriate next step in diagnosis?

A) HIV testing
B) Fungal culture
C) Herpes culture
D) Human papillomavirus culture
E) Colposcopy

3. Six hours after beginning dicloxacillin therapy for cellulitis of the right lower extremity, a 37-year-old woman comes to the emergency department because of a 3-hour history of pruritic rash, dyspnea, and mild cough. She has a history of childhood asthma but no history of adverse reactions to medication. Six months ago, she had streptococcal pharyngitis treated with penicillin. She has otherwise been healthy. Her temperature is 37.1°C (98.8°F), pulse is 110/min, respirations are 22/min, and blood pressure is 86/64 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 92%. Examination of the skin shows 1- to 2-cm areas of pink urticaria over the trunk and upper and lower extremities and a 3-cm, erythematous, mildly tender area over the distal right lower extremity. Air movement is poor, and inspiratory and expiratory wheezes are heard on auscultation. Her hemoglobin concentration is 13.1 g/dL, and leukocyte count is 9500/$mm^3$. A chest x-ray shows no abnormalities. In addition to discontinuing dicloxacillin therapy, which of the following is the most appropriate next step in pharmacotherapy?

A) Epinephrine
B) Loratadine
C) Prednisone
D) Stanozolol

4. A previously healthy 47-year-old man comes to the physician because of a 3-month history of progressively worsening fatigue; he has had a 14-kg (30-lb) weight loss during this period. Vital signs are within normal limits. The spleen tip is palpated 8 cm below the left costal margin. Examination shows no other abnormalities. His hematocrit is 40%, leukocyte count is 36,000/mm$^3$, and platelet count is 70,000/mm$^3$. Bone marrow aspirate and biopsy show 100% cellularity with an increase in all hematopoietic cell lines. The presence of which of the following is most likely to confirm the diagnosis?

A) BCR/ABL fusion protein
B) *BRCA1* oncogene
C) P-glycoprotein
D) p53 Suppressor gene
E) t (8,16) Translocation

5. A 42-year-old woman comes to the physician for a follow-up examination. She has had paraplegia since being involved in a motor vehicle collision 12 years ago. She has had recurrent urinary tract infections over the past 10 years; the most recent infection occurred 3 months ago. Examination shows no abnormalities other than paraplegia. Test of the stool for occult blood is negative. Laboratory studies show:

| | |
|---|---|
| Hemoglobin | 8.7 g/dL |
| Mean corpuscular volume | 90 $\mu m^3$ |
| Leukocyte count | 6300/$mm^3$ |
| Platelet count | 160,000/$mm^3$ |

Which of the following is the most likely cause of this patient's decreased hemoglobin concentration?

A) Cytokine-induced erythropoietin deficiency
B) Folic acid deficiency
C) Glucose 6-phosphate dehydrogenase deficiency
D) Iron deficiency
E) Myelodysplasia

6. A 32-year-old African American man with iron deficiency anemia comes to the office for a follow-up examination. He has adhered to his medication regimen of ferrous sulfate, which he began 6 months ago. He has not had bloody vomiting or blood in his stool or urine. He has smoked one-half pack of cigarettes daily for 10 years. He drinks six 12-oz beers on the weekends. Vital signs are within normal limits. There is no conjunctival pallor or icterus. The remainder of the examination shows no abnormalities. Laboratory studies show:

| | **6 Months Ago** | **Today** |
|---|---|---|
| Hemoglobin (g/dL) | 10.7 | 10.9 |
| Hematocrit (%) | 32 | 33 |
| Erythrocyte count (/mm$^3$) | 5 million | 4.8 million |
| Mean corpuscular volume (μm$^3$) | 72 | 72 |
| Leukocyte count (/mm$^3$) | 4100 | 4300 |
| Platelet count (/mm$^3$) | 225,000 | 205,000 |
| Red cell distribution width (%) (N=13–15) | 13.8 | 13.5 |
| Blood smear | microcytosis and target cells | microcytosis and target cells |
| Serum | | |
| Iron (μg/dL) | 60 | 75 |
| Transferrin saturation (%) (N=20–50) | 25 | 35 |

Which of the following is the most likely diagnosis?

A ) Celiac disease
B ) Myelodysplastic syndrome
C ) Occult gastrointestinal bleeding
D ) Pure red blood cell aplasia
E ) Thalassemia

7. A 23-year-old man with sickle cell disease comes to the emergency department because of low back pain for 12 hours. During this time, he also has had nausea and decreased intake of solids and liquids. He rates the pain as an 8 on a 10-point scale. He has been taking ibuprofen every 8 hours with no relief of the pain. He appears to be in distress. Vital signs are within normal limits. Examination shows pale conjunctivae. His hemoglobin concentration is 7.2 g/dL, serum urea nitrogen concentration is 32 mg/dL, and serum creatinine concentration is 2.6 mg/dL. In addition to administration of oxygen and 0.9% saline, which of the following is the most appropriate next step in pharmacotherapy?

- A) Oral celecoxib
- B) Oral oxycodone
- C) Intramuscular ketorolac
- D) Intravenous meperidine
- E) Intravenous morphine

8. A 72-year-old man comes for a routine examination. He has a 20-year history of poorly controlled hypertension with frequent diastolic blood pressures of 105 mm Hg. He smokes 15 cigarettes and drinks three glasses of wine daily. His father died of a cerebral infarction at the age of 58 years. His blood pressure now is 150/105 mm Hg; the remainder of the examination shows no abnormalities. His fasting serum glucose concentration is 180 mg/dL, and serum LDL-cholesterol concentration is 170 mg/dL. Which of the following is the greatest risk factor for cerebral infarction in this patient?

A) Abnormal serum lipid concentrations
B) Family history of cerebral infarction
C) Hypertension
D) Increase in serum glucose concentration
E) Smoking history

9. A 77-year-old woman with severe degenerative arthritis comes to the office because of a 2-week history of ringing and pain in her ears. She has no other history of serious illness. She has been taking aspirin and acetaminophen with minimal relief of pain. Vital signs are within normal limits. Examination of the shoulders and knees shows no swelling; active range of motion elicits pain bilaterally. Which of the following laboratory findings is most likely in this patient?

| | $Na^+$ (mEq/L) | $K^+$ (mEq/L) | $Cl^-$ (mEq/L) | $HCO_3^-$ (mEq/L) | pH |
|---|---|---|---|---|---|
| A) | 112 | 3.4 | 78 | 24 | 7.40 |
| B) | 132 | 3.2 | 104 | 18 | 7.34 |
| C) | 132 | 4.6 | 94 | 18 | 7.38 |
| D) | 140 | 5.5 | 106 | 20 | 7.24 |
| E) | 154 | 3.8 | 118 | 25 | 7.42 |
| F) | 154 | 5.5 | 110 | 20 | 7.24 |

10. A 23-year-old woman comes to the physician because of a lesion on her lip that appeared 1 day ago. She reports feeling a tingling and burning sensation in that same area for the past 4 days. Examination of the lower lip shows a grouping of discrete, clear, fluid-filled vesicles that are 5 mm in diameter. No other abnormalities are noted. Which of the following is the most likely diagnosis?

- A) Chickenpox
- B) Erythema multiforme
- C) Erythema nodosum
- D) Herpes simplex
- E) Herpes zoster

11. An otherwise healthy 42-year-old man comes to the physician because of a 3-month history of a progressive rash over his arms and back. The rash is mildly itchy. His temperature is 36.7°C (98°F), pulse is 72/min, respirations are 16/min, and blood pressure is 115/80 mm Hg. A photograph of the rash is shown. Which of the following is the most likely diagnosis?

A) Atopic dermatitis
B) Bullous pemphigoid
C) Dysplastic nevi
D) Lichen simplex chronicus
E) Pityriasis rosea
F) Psoriasis
G) Tinea corporis

12. A 57-year-old man comes to the physician because of a 12-week history of left shoulder pain that radiates to the middle of his left upper arm. He is awakened at night by pain when he rolls onto the affected shoulder. He does not recall any recent history of trauma. He has no history of serious illness and takes no medications. Examination shows no muscular atrophy. There is tenderness to palpation immediately lateral to the acromion. Active abduction of the left shoulder is limited to 120 degrees without resistance and difficult with minimal resistance. Peripheral pulses are normal. Neurologic examination shows no focal findings. X-rays of the left shoulder show no abnormalities. Which of the following is the most likely diagnosis?

A) Accessory nerve palsy
B) Acromioclavicular separation
C) Occult fracture of the proximal humerus
D) Rotator cuff tendinitis
E) Suprascapular nerve palsy

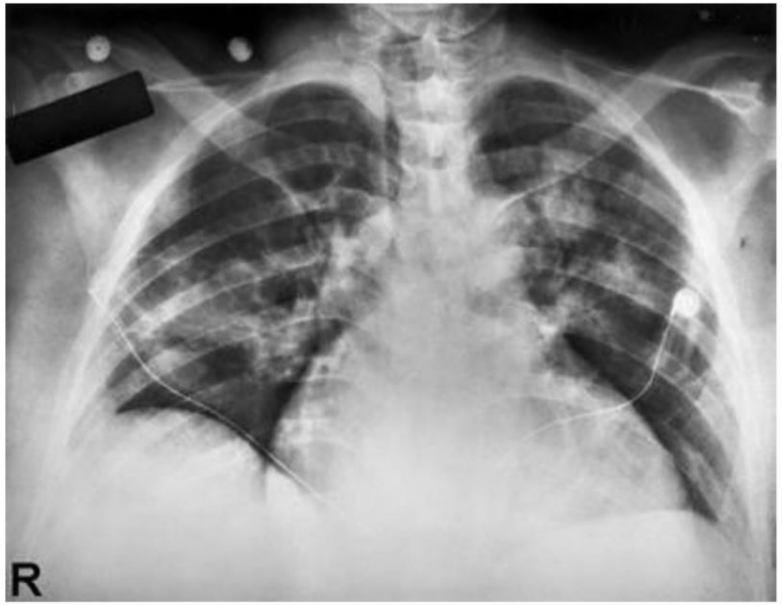
13. A 67-year-old man comes to the office because of a 3-day history of moderate pain and swelling of his right knee. He has not sustained any recent trauma. He has hypertension and hypothyroidism. His medications are lisinopril and levothyroxine. He is 178 cm (5 ft 10 in) tall and weighs 91 kg (200 lb); BMI is 29 kg/$m^2$. His temperature is 37.8°C (100°F), pulse is 84/min, and blood pressure is 135/88 mm Hg. Examination of the right knee shows erythema, warmth, and moderate effusion; there is tenderness to palpation over the lateral joint line. Active range of motion of the right knee is limited by pain. An x-ray of the right knee shows chondrocalcinosis. Which of the following is the most likely mechanism of this patient's condition?

A) Deposition of calcium pyrophosphate dihydrate crystals
B) Deposition of monosodium urate crystals
C) Erosive inflammation associated with the loss of articular cartilage
D) Infection of the joint space with gram-negative rods
E) Infection of the joint space with gram-positive cocci

14. A 57-year-old man comes to the physician because of a 1-week history of a painful mass on his right arm that developed after he lifted several heavy boxes. He is right-hand dominant. There is no other history of trauma to the arm. He has a 5-year history of chronic right shoulder pain but is otherwise asymptomatic. He has smoked two packs of cigarettes daily for 30 years. His temperature is 37.2°C (99°F). A 2 × 3-cm, tender, mobile mass is palpated in the anterior aspect of the right upper extremity. There is pain with forward flexion of the right shoulder. Active flexion of the right elbow is limited by pain; elbow extension strength is full and painless. Muscle strength and sensation in the right wrist and hand are intact. Radial and ulnar pulses are 2+. Plain x-rays of the right upper extremity show a vague, anterior soft-tissue shadow. Which of the following is the most likely diagnosis?

A) Abscess
B) Arteriovenous malformation
C) Biceps tendon rupture
D) Lipoma
E) Malignant fibrous histiocytoma

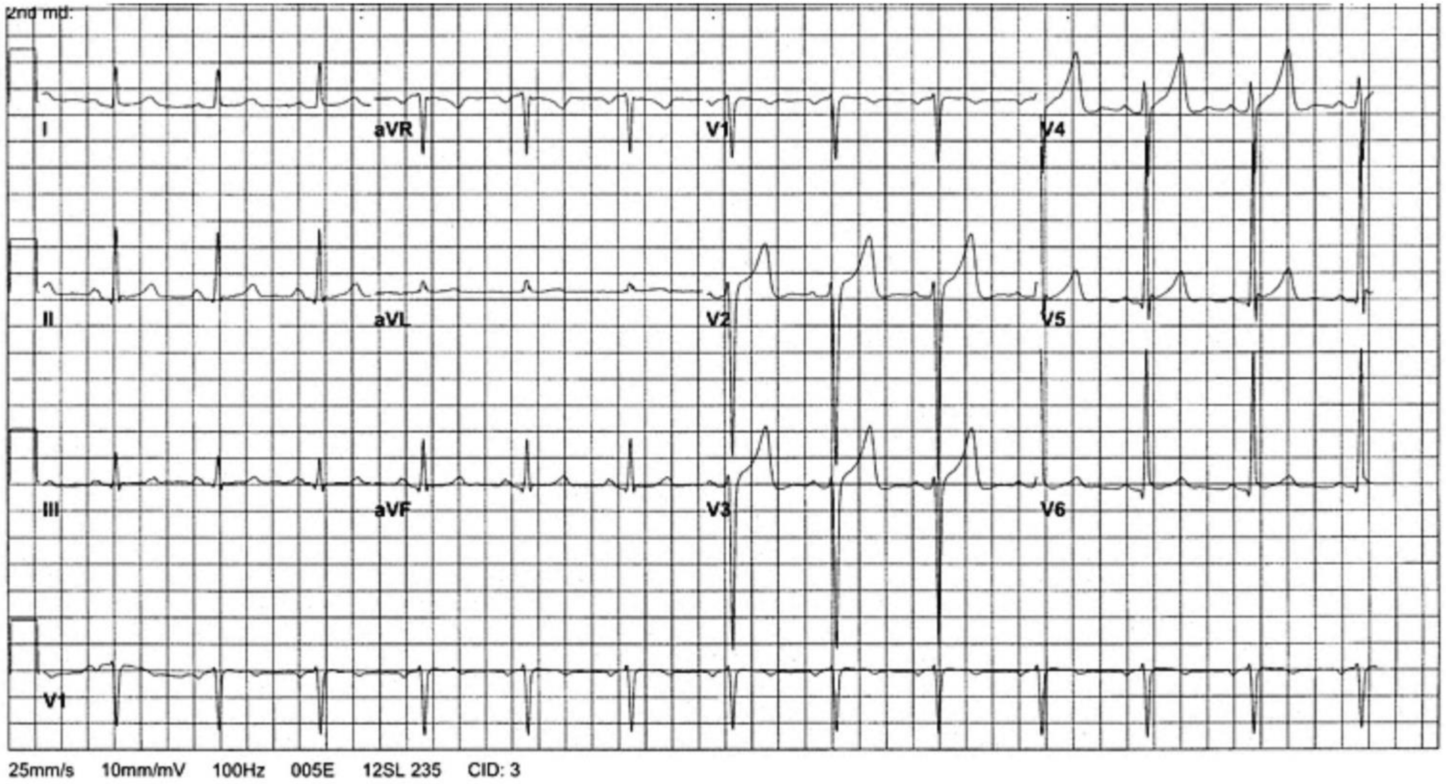
15. A 54-year-old man is admitted to the hospital because of a 1-week history of temperatures to 38.3°C (101°F), chills, progressive shortness of breath, and pain and swelling of his right knee. He has a 20-year history of poorly controlled type 2 diabetes mellitus treated with metformin and insulin. His temperature is 38.7°C (101.6°F), pulse is 100/min, respirations are 20/min, and blood pressure is 110/60 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 92%. Cardiac examination discloses a grade 3/6 holosystolic murmur heard best at the apex. Scattered crackles are heard throughout the lung fields. The right knee is swollen and tender to palpation. There is a 2-cm ulcer over the bottom of the right foot; no surrounding erythema or soft-tissue swelling is noted. When asked about the ulcer, the patient says it is not painful and that it has been present for 5 months. A chest x-ray is shown. In addition to aspiration of the right knee joint, which of the following is the most appropriate next step in diagnosis?




A) Blood cultures
B) Bone biopsy
C) Culture of the ulcer tissue
D) Lung biopsy
E) Measurement of serum ACE activity
F) Transthoracic echocardiography

16. A 78-year-old woman is admitted to the hospital 2 hours after the sudden onset of shortness of breath, palpitations, and dizziness. She has hypertension and heart failure with diastolic dysfunction. Her medications are lisinopril, metoprolol, hydrochlorothiazide, and aspirin. Her temperature is 36.7°C (98.1°F), pulse is 150/min and irregularly irregular, respirations are 28/min, and blood pressure is 82/50 mm Hg. Pulse oximetry on 50% oxygen by nonrebreather face mask shows an oxygen saturation of 88%. Crackles are heard one-third of the way up the lung fields. There is inspiratory and expiratory wheezing. No peripheral edema is noted. An ECG shows atrial fibrillation with a ventricular rate of 160/min. Which of the following is the most appropriate next step in management?

A) Administration of adenosine
B) Administration of albuterol
C) Administration of amiodarone
D) Administration of diltiazem
E) Administration of esmolol
F) Electrical cardioversion
G) Placement of a pacemaker




17. A 72-year-old woman is admitted to the hospital because of a 3-week history of progressive shortness of breath that increases with exertion. She has a 20-year history of hypertension poorly controlled with hydrochlorothiazide. She takes no other medications. She has smoked one pack of cigarettes daily for 55 years. She is in mild distress. Her temperature is 36.7°C (98°F), pulse is 90/min and regular, respirations are 20/min, and blood pressure is 180/100 mm Hg. There is no jugular venous distention or peripheral edema. Crackles are heard over the lower lung fields bilaterally. On cardiac examination, an $S_3$ is heard; there are no murmurs. Laboratory studies show:

| | |
|---|---|
| Hemoglobin | 14 g/dL |
| Leukocyte count | 8000/mm$^3$ |
| Serum | |
| Urea nitrogen | 30 mg/dL |


17. A 72-year-old woman is admitted to the hospital because of a 3-week history of progressive shortness of breath that increases with exertion. She has a 20-year history of hypertension poorly controlled with hydrochlorothiazide. She takes no other medications. She has smoked one pack of cigarettes daily for 55 years. She is in mild distress. Her temperature is 36.7°C (98°F), pulse is 90/min and regular, respirations are 20/min, and blood pressure is 180/100 mm Hg. There is no jugular venous distention or peripheral edema. Crackles are heard over the lower lung fields bilaterally. On cardiac examination, an $S_3$ is heard; there are no murmurs. Laboratory studies show:

| | |
|---|---|
| Hemoglobin | 14 g/dL |
| Leukocyte count | 8000/$mm^3$ |
| Serum | |
| Urea nitrogen | 30 mg/dL |
| Creatinine | 2 mg/dL |
| Troponin I | <0.01 ng/mL (N<0.4) |

An ECG is shown. Echocardiography shows normal heart valves and a left ventricular ejection fraction of 58%. Which of the following is the most likely diagnosis?

A) Biventricular systolic dysfunction
B) Diastolic dysfunction
C) High-output heart failure
D) Left ventricular systolic dysfunction
E) Right ventricular systolic dysfunction

18. A 57-year-old woman is brought to the emergency department 8 hours after a 2-hour episode of shortness of breath and substernal chest pain that radiated to her left shoulder. She rates the episode of pain as a 6 on a 10-point scale. She has a 30-year history of hypertension well controlled with lisinopril and a 20-year history of hypercholesterolemia treated with atorvastatin. She has smoked one pack of cigarettes daily for 40 years. Her pulse is 90/min, respirations are 18/min, and blood pressure is 160/92 mm Hg. Bilateral basilar crackles are heard. Cardiac examination shows no abnormalities except for an $S_4$ gallop. Her serum troponin I concentration is 3.6 ng/mL (N<0.4). An x-ray of the chest shows moderate cardiomegaly. An ECG shows T-wave inversion in the anterior and lateral precordial leads. Which of the following findings is the best evidence of an acute myocardial infarction in this patient?

A) Cardiomegaly on x-ray of the chest
B) Chest pain pattern
C) Increased serum troponin I concentration
D) Presence of crackles
E) Presence of an $S_4$ gallop
F) T-wave inversion in the anterior and lateral precordial leads on ECG

19. A 37-year-old woman comes to the physician because of progressive shortness of breath and blood-tinged sputum over the past 7 days. Examination shows jugular venous distention. There is an increased $S_1$ with a high-pitched extra heart sound after $S_2$. A grade 1/6, low-pitched diastolic murmur is heard at the apex radiating to the anterior axillary line. Which of the following is the most likely cause of these findings?

A) Aortic insufficiency
B) Aortic stenosis
C) Atrial septal defect
D) Coarctation of the aorta
E) Mitral insufficiency
F) Mitral stenosis
G) Pulmonic stenosis

20. A hospitalized 57-year-old man has had the gradual onset of edema of his left lower extremity during the past 4 hours. He underwent prostatectomy for carcinoma 48 hours ago. He has not had pain in his extremities. Pneumatic compression stockings were applied to both lower extremities during the operation. Prophylactic low-molecular-weight heparin therapy was begun 24 hours postoperatively. His only other medication is morphine. He is alert. His temperature is 38.4°C (101.1°F), pulse is 85/min, respirations are 13/min, and blood pressure is 145/80 mm Hg. Examination of the left lower extremity shows edema from the knee to the dorsum of the foot. Peripheral pulses are normal. The remainder of the examination shows no abnormalities. Venous duplex ultrasonography of the left lower extremity shows occlusion of the popliteal vein and the distal superficial femoral vein. Which of the following is the most appropriate next step in management?

A) Clopidogrel therapy
B) Compression therapy
C) Dipyridamole therapy
D) Heparin therapy
E) Thrombectomy
F) Tissue plasminogen activator therapy
G) Venous ligation
H) Venous resection
I) Warfarin therapy

21. A 57-year-old man is brought to the emergency department by his wife because of a 2-day history of progressive shortness of breath, fever, and cough productive of clear sputum. Initially, his shortness of breath occurred with exertion; now, it occurs at rest. He received the diagnosis of non-small cell carcinoma of the lung 1 year ago; since then, he has had stable dyspnea with exertion. He has received chemotherapy and radiation therapy; his last course of therapy was 2 months ago. He takes no medications. On arrival, he is in mild respiratory distress. His temperature is 38.9°C (102°F), pulse is 104/min and regular, respirations are 26/min, and blood pressure is 146/74 mm Hg. Crackles are heard at the right lung base. His leukocyte count is 16,800/$mm^3$ (75% segmented neutrophils, 5% bands, 2% eosinophils, 1% basophils, 12% lymphocytes, and 5% monocytes). A chest x-ray shows an increased right perihilar mass and a new right middle lobe infiltrate. Which of the following is the most likely cause of this patient's pneumonia?

A) Airway obstruction secondary to carcinoma
B) Aspiration from recurrent laryngeal nerve compression
C) Immunosuppression from chemotherapy
D) Immunosuppression from radiation therapy
E) Impaired mucous clearance

22. An 82-year-old woman comes to the physician because of a 2-week history of cough productive of copious amounts of green sputum. She has had several similar episodes over the past 5 years. She had several episodes of "walking pneumonia" as a child. She does not smoke and has no history of asthma or exposure to sick contacts. She is afebrile. The lungs are clear to auscultation. Examination shows clubbing of the fingers. An x-ray of the chest shows streaky linear bilateral basilar scarring. Which of the following is the most likely diagnosis?

A) $\alpha_1$-Antitrypsin deficiency
B) Asthma
C) Bronchiectasis
D) Cystic fibrosis
E) Gastroesophageal reflux
F) Hereditary angioedema
G) *Mycobacterium tuberculosis* infection
H) Postnasal discharge
I ) Sarcoidosis

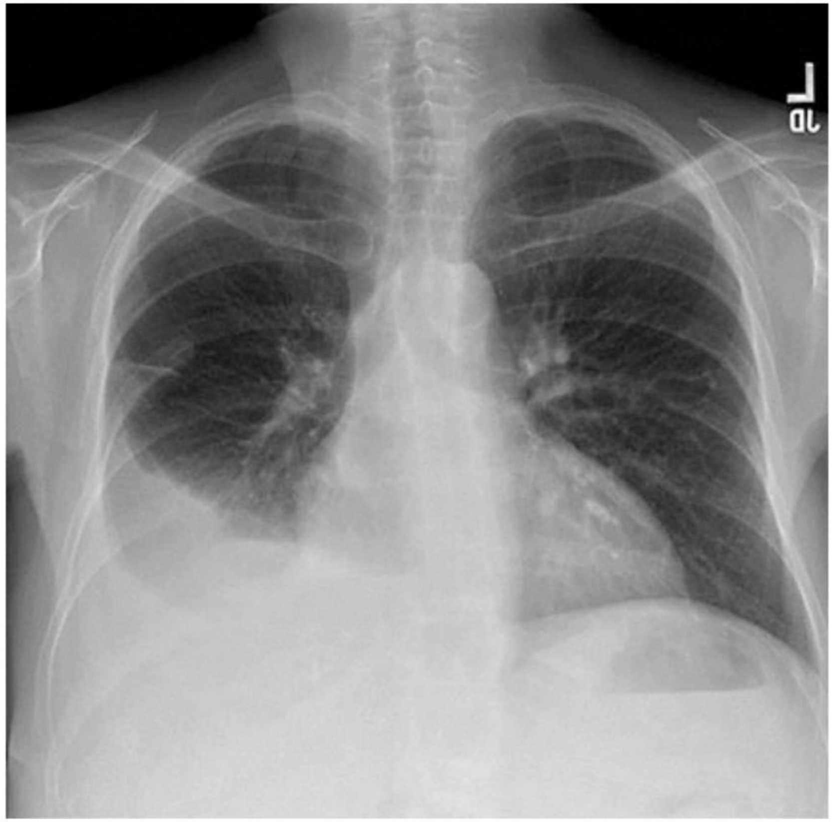
23. A hospitalized 18-year-old woman with newly diagnosed asthma is evaluated 12 hours before discharge. On admission 2 days ago, she reported a 4-month history of intermittent shortness of breath, nonproductive cough, and wheezing. Her symptoms initially occurred only with exercise; 1 month ago, they began to occur daily with minimal exertion. During this time, her wheezing also awakened her several times weekly. She has no other history of serious illness and takes no medications. Since admission, she has received nebulized albuterol and oral prednisone. She is alert, has no difficulty speaking, and reports no shortness of breath. Her pulse is 88/min, respirations are 22/min, and blood pressure is 128/82 mm Hg. Pulse oximetry on 2 L/min of oxygen by nasal cannula shows an oxygen saturation of 98%. She is not using accessory muscles of respiration. There are mild, scattered wheezes on expiration. Upon discharge, the patient is switched from nebulized albuterol. In addition to completing the course of prednisone begun in the hospital, which of the following is the most appropriate additional pharmacotherapy?

A) Inhaled cromolyn
B) Inhaled fluticasone
C) Inhaled salmeterol
D) Oral cetirizine
E) Oral montelukast

24. A 57-year-old woman comes to the physician because of a 6-month history of fatigue and shortness of breath with exertion. She feels winded after climbing two flights of stairs. She has no history of serious illness. Menopause occurred 9 years ago. She was receiving hormone replacement therapy but discontinued it 4 years ago because of concern about breast cancer. She does not smoke. Her pulse is 88/min, and blood pressure is 130/85 mm Hg. Crackles are heard at the lung bases. The remainder of the examination shows no abnormalities. An x-ray of the chest shows interstitial markings but no effusions. Pulmonary function tests show a decreased diffusion capacity, a decreased FVC, and an increased $FEV_1$:FVC ratio consistent with a restrictive pattern. Which of the following is the most appropriate next step in diagnosis?

A) PPD skin test
B) Serum rheumatoid factor assay
C) Exercise stress test
D) High-resolution CT scan of the chest
E) Ventilation-perfusion lung scans

25. A 78-year-old man is admitted to the hospital because of a 6-week history of progressive shortness of breath. He was diagnosed with colon cancer at age 55 years. He has congestive heart failure, mitral regurgitation, and chronic obstructive pulmonary disease. His medications are lisinopril, atenolol, hydrochlorothiazide, tiotropium, and supplemental oxygen therapy (2 L/min by nasal cannula). He appears anxious and is in severe respiratory distress. He uses accessory muscles of respiration. His temperature is 37.2°C (99.0°F), pulse is 111/min, respirations are 22/min, and blood pressure is 134/88 mm Hg. Pulse oximetry on 2 L/min of oxygen by nasal cannula shows an oxygen saturation of 86%. Physical examination shows mild jugular venous distention and hypertrophy of the sternocleidomastoid muscles. Pulmonary examination discloses decreased breath sounds in the right lower lobe; there is dullness to percussion. Cardiac examination discloses a grade 3/6 holosystolic murmur heard best at the left sternal border. There is 2+ edema of the lower extremities to the knees. A chest x-ray is shown. Thoracentesis is performed. Results of laboratory studies are shown:




| | |
|---|---|
| Hemoglobin | 11 mg/dL |
| Leukocyte count | 11,000/$mm^3$ |
| Segmented neutrophils | 64% |
| Lymphocytes | 24% |
| Monocytes | 12% |
| Serum | |
| Protein | 5.2 g/dL |
| Lactate dehydrogenase | 230 U/L |
| Pleural fluid | |
| pH | 7.38 |
| RBC | 350/$mm^3$ |
| WBC | 130/$mm^3$ |
| Protein | 1.2 g/dL |
| Lactate dehydrogenase | 85 U/L |

Which of the following is the most appropriate initial pharmacotherapy?

A) Ceftriaxone
B) Enoxaparin

distress. He uses accessory muscles of respiration. His temperature is [illegible], pulse is [illegible]/min, respirations are 22/min, and blood pressure is 134/88 mm Hg. Pulse oximetry on 2 L/min of oxygen by nasal cannula shows an oxygen saturation of 86%. Physical examination shows mild jugular venous distention and hypertrophy of the sternocleidomastoid muscles. Pulmonary examination discloses decreased breath sounds in the right lower lobe; there is dullness to percussion. Cardiac examination discloses a grade 3/6 holosystolic murmur heard best at the left sternal border. There is 2+ edema of the lower extremities to the knees. A chest x-ray is shown. Thoracentesis is performed. Results of laboratory studies are shown:

| | |
|---|---|
| Hemoglobin | 11 mg/dL |
| Leukocyte count | $11,000/mm^3$ |
| Segmented neutrophils | 64% |
| Lymphocytes | 24% |
| Monocytes | 12% |
| Serum | |
| Protein | 5.2 g/dL |
| Lactate dehydrogenase | 230 U/L |
| Pleural fluid | |
| pH | 7.38 |
| RBC | $350/mm^3$ |
| WBC | $130/mm^3$ |
| Protein | 1.2 g/dL |
| Lactate dehydrogenase | 85 U/L |

Which of the following is the most appropriate initial pharmacotherapy?

A) Ceftriaxone
B) Enoxaparin
C) Furosemide
D) Methylprednisolone
E) Morphine

26. A 57-year-old woman comes to the physician because of a 10-month history of severe fatigue. She awakens four to five times nightly to void. She now takes daily naps and frequently falls asleep while watching television. She has noted swelling of her legs and occasional palpitations during the past 6 months. She has not had fever, night sweats, or change in bowel movements. She has had an 18-kg (40-lb) weight gain during the past year. She is 160 cm (5 ft 3 in) tall and now weighs 145 kg (320 lb); BMI is 57 kg/m$^2$. Her pulse is 96/min, respirations are 20/min, and blood pressure is 160/95 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 92%. An x-ray of the chest shows no abnormalities. An ECG shows sinus tachycardia and P pulmonale. Which of the following is the most likely cause of this patient's nocturia?

A ) Chronic obstructive pulmonary disease
B ) Cystitis
C ) Hypothyroidism
D ) Narcolepsy
E ) Obesity-hypoventilation syndrome

27. A 37-year-old woman comes to the physician because of a 1-month history of midabdominal pain after meals. She has a 10-year history of Crohn disease and has recurrent episodes of abdominal pain. She underwent resection of the terminal ileum and cecum 5 years ago and two jejunal strictureplasties 1 year ago. Current medications include an aminosalicylate. Her temperature is 38.4°C (101.1°F). Abdominal examination shows moderate midsupraumbilical tenderness. There is a well-healed surgical scar. Test of the stool for occult blood is negative. An upper gastrointestinal series with contrast shows a 10-cm segment of ileum with irregular mucosa and no strictures. Which of the following is the most appropriate next step in management?

A) Bowel rest
B) Cyclosporine therapy
C) Infliximab therapy
D) Prednisone therapy
E) Surgical exploration

28. A previously healthy 37-year-old man comes to the physician because of a 1-year history of intermittent burning midepigastric pain and a 2-month history of diarrhea. The pain is relieved by over-the-counter ranitidine. He passes 8 to 10 large, watery, nonbloody stools daily. Examination shows no abnormalities. Test of the stool for occult blood is positive. Upper endoscopy shows multiple duodenal ulcers. Which of the following is the most likely diagnosis?

- A) Bowel lymphoma
- B) Gastrinoma
- C) Parathyroid adenoma
- D) Pheochromocytoma
- E) VIPoma

29. A previously healthy 23-year-old woman, who is a graduate student, comes to the office because of a 4-month history of intermittent diarrhea characterized by copious loose brown stools and moderate abdominal cramps. The cramps are relieved by defecation. She has not had fever, nausea, vomiting, or weight loss. She takes ginger extract and a daily multivitamin. She smokes three to five cigarettes daily. She drinks one to three beers on special occasions. She does not use illicit drugs. She has not traveled recently. She is 168 cm (5 ft 6 in) tall and weighs 63 kg (140 lb); BMI is 23 kg/m$^2$. Her temperature is 37.2°C (99°F), pulse is 72/min, and blood pressure is 112/68 mm Hg. Examination, including examination of the oropharynx and abdomen, shows no abnormalities. Results of laboratory studies are within the reference ranges. Which of the following mechanisms is most likely responsible for these findings?

A) Bacteria overgrowth
B) Endomysial antibody formation
C) Medication adverse effect
D) Smooth muscle hypersensitivity
E) Stimulant laxative use

30. An 82-year-old woman is brought to the emergency department by her family because of progressive loss of appetite and yellowing of the skin over the past 2 months. She has had a 4.5-kg (10-lb) weight loss during this period but says that she feels well. Her temperature is 37°C (98.6°F), pulse is 100/min, respirations are 12/min, and blood pressure is 110/80 mm Hg while supine and 80/60 mm Hg while sitting. The skin and sclerae appear yellow and dry, and there is tenting over the forehead. The liver edge is palpated 10 cm below the right costal margin. Examination shows no other abnormalities. Laboratory studies show:

| | |
|---|---|
| Hemoglobin | 10 g/dL |
| Serum | |
| Bilirubin, total | 3.4 mg/dL |
| Direct | 2.0 mg/dL |
| Alkaline phosphatase | 500 U/L |
| AST | 75 U/L |

Which of the following is the most appropriate next step in diagnosis?

A) Barium enema
B) Cholecystography
C) CT scan of the abdomen
D) Esophagogastroduodenoscopy

31. A 47-year-old man with a history of alcoholism is brought to the emergency department because of severe abdominal pain, mild nausea, and two episodes of vomiting during the past 16 hours. The pain is located in the epigastrium and radiates straight through to the back. He had a similar episode of pain 2 years ago that was treated with fluid hydration and bowel rest during a 2-day hospital stay, but he left against medical advice. Vital signs are within normal limits except for a temperature of 38.7°C (101.7°F). Abdominal examination shows slight distention and mild epigastric tenderness. The liver is nontender, and the edge is palpable at the costal margin. The spleen is not palpable. Bowel sounds are hypoactive. Test of the stool for occult blood is negative. Insertion of a nasogastric tube yields green-tinged fluid. His leukocyte count is 18,000/mm$^3$, and serum amylase activity is 350 U/L. Liver function tests show no abnormalities. Which of the following is the most appropriate next step in management?

A) X-rays of the abdomen
B) CT scan of the abdomen
C) HIDA scan
D) MRI of the abdomen
E) Endoscopy
F) Endoscopic retrograde cholangiopancreatography
G) Percutaneous aspiration

32. A 24-year-old man comes to the office because his fiancé had a urinalysis 2 days ago that confirmed the presence of *Trichomonas vaginalis*. They are sexually active and use condoms inconsistently. He has not had urinary frequency or urgency, painful urination, or dark urine. He has no history of serious illness, takes no medications, and has no known drug allergies. Vital signs are within normal limits. Examination shows no abnormalities. Which of the following is the most appropriate next step in management?

A) Amoxicillin therapy
B) Follow-up examination if he becomes symptomatic
C) Metronidazole therapy
D) Urethral swab
E) Urinalysis

33. A 72-year-old woman with multiple myeloma is brought to the physician because of a 1-day history of shortness of breath and fatigue. Her respirations are 20/min, and blood pressure is 120/60 mm Hg with no orthostatic changes. The lungs are clear to auscultation. Cardiac examination shows decreased heart sounds. The remainder of the examination shows no abnormalities. Laboratory studies show:

| | |
|---|---|
| Serum | |
| Urea nitrogen | 36 mg/dL |
| Creatinine | 5.1 mg/dL |
| Urine | |
| Protein | 3+ |
| RBC | 0/hpf |
| WBC | 0/hpf |
| Fractional excretion of sodium | >1% |
| Postvoid residual volume | 50 mL |

A 24-hour urine collection shows 3.5 g of protein and a creatinine clearance of 10 mL/min. Which of the following is the most likely cause of these laboratory findings?

A) Bladder outlet obstruction
B) Congestive heart failure
C) Interstitial nephritis
D) Renal parenchymal disease
E) Renovascular disease

34. A 27-year-old man comes to the physician because of a 3-month history of progressive generalized swelling; he has had a 10-kg (22-lb) weight gain during this period. He has a 10-year history of heroin use. He has smoked one pack of cigarettes daily for 13 years. He had hepatitis B 10 years ago and mitral valve endocarditis 2 years ago. An HIV antibody assay was negative 1 month ago. His temperature is 37.4°C (99.3°F), pulse is 100/min, and blood pressure is 150/110 mm Hg. Examination shows no jugular venous distention. There is bilateral basilar dullness to percussion, and breath sounds are decreased. A grade 2/6 holosystolic murmur is heard at the apex that radiates to the axilla. Examination shows a protuberant abdomen with a fluid wave. There is 3+ pitting edema to the lower abdominal wall. The scrotum is markedly edematous. Laboratory studies show:

| | | | |
|---|---|---|---|
| Serum | | Serum | |
| Urea nitrogen | 38 mg/dL | Hepatitis B surface antigen | negative |
| Glucose | 154 mg/dL | Hepatitis B surface antibody | positive |
| Creatinine | 3 mg/dL | Hepatitis C antibody | negative |
| Albumin | 2 g/dL | Urine | |
| Cholesterol | 280 mg/dL | Protein | 4+ |
| Total bilirubin | 1 mg/dL | RBC | 50/hpf |
| AST | 250 U/L | WBC | 8/hpf |
| ALT | 35 U/L | RBC casts | none |
| C3 | 57 mg/dL (N=55–120) | Granular casts | occasional |
| C4 | 20 mg/dL (N=20–50) | Leukocyte esterase | 2+ |

The most likely explanation for these findings is nephropathy induced by which of the following?

- A) Cholesterol
- B) Cryoglobulin
- C) Diabetes mellitus
- D) Heroin
- E) HIV infection

35. One day after undergoing chemotherapy for acute lymphocytic leukemia, a 38-year-old man has an increase in his serum urea nitrogen concentration from 20 mg/dL to 60 mg/dL. Examination of urine sediment shows crystals. Treatment with which of the following drugs prior to chemotherapy is most likely to have prevented this acute renal failure?

- A) Allopurinol
- B) Amoxicillin
- C) Furosemide
- D) Indomethacin
- E) Prednisone

36. Two days after the onset of menses, a 24-year-old woman develops diffuse, migrating pain of her wrists, elbows, knees, and ankles and pustules on her extremities. Examination shows slightly tender pustules on the upper and lower extremities. A photograph of the right fingers is shown. There is moderate swelling of the right knee. Which of the following drugs is most likely to be effective?

A) Acyclovir
B) Ceftriaxone
C) Sulfamethoxazole
D) Tetracycline

37. A 62-year-old man comes to the emergency department because of a 12-hour history of pelvic discomfort and an inability to void. During the past year, he has had progressive difficulty initiating a urinary stream and dribbles at the end of urination. He has not received medical care for 20 years. He has no history of serious illness and takes no medications. He is 173 cm (5 ft 8 in) tall and weighs 86 kg (190 lb); BMI is 29 kg/m$^2$. His temperature is 37.1°C (98.8°F), pulse is 80/min, respirations are 16/min, and blood pressure is 150/90 mm Hg. There is an $S_4$. Abdominal examination shows lower quadrant tenderness. There is 2+ pitting edema. Laboratory studies show:

| | |
|---|---|
| Hemoglobin $A_{1c}$ | 8.9% |
| Serum | |
| $Na^+$ | 140 mEq/L |
| $K^+$ | 5.8 mEq/L |
| $Cl^-$ | 106 mEq/L |
| $HCO_3^-$ | 19 mEq/L |
| Urea nitrogen | 52 mg/dL |
| Glucose | 250 mg/dL |
| Creatinine | 2.9 mg/dL |

Which of the following is the most likely cause of this patient's increased serum creatinine concentration?

A) Bladder outlet obstruction
B) Crescent formation in the glomeruli
C) Necrosis of the renal tubules
D) Protein deposition in the kidneys
E) Stenosis of the renal arteries

38. An asymptomatic 67-year-old woman comes to the physician for a routine follow-up visit. She has a 7-year history of type 2 diabetes mellitus treated with diet. Her fingerstick blood glucose concentrations have ranged between 100 mg/dL and 140 mg/dL during the past 6 months. Her pulse is 80/min and regular, and blood pressure is 118/82 mm Hg. Examination shows no abnormalities. Laboratory studies show:

| | |
|---|---|
| Hemoglobin | 12 g/dL |
| Hemoglobin $A_{1c}$ | 7% |
| Serum | |
| Urea nitrogen | 18 mg/dL |
| Glucose (fasting) | 115 mg/dL |
| Creatinine | 0.8 mg/dL |
| Total cholesterol | 275 mg/dL |
| LDL-cholesterol | 190 mg/dL |
| HDL-cholesterol | 35 mg/dL |
| Urine microalbumin | negative |

Which of the following is the most appropriate intervention to reduce long-term morbidity and mortality in this patient?

A) Decrease in systolic blood pressure to 100 mm Hg
B) Reduction of serum LDL-cholesterol concentration
C) ACE inhibitor therapy
D) Folic acid supplementation
E) Glyburide therapy

39. A 47-year-old woman with end-stage renal disease comes to the office because of a 6-week history of pain in her shoulders and shins. She has received hemodialysis three times weekly for 12 years; she last received hemodialysis 2 days ago. Vital signs are within normal limits. Examination shows a thrill at the site of an arteriovenous fistula in the left forearm and trace pedal edema. X-rays of the clavicle and tibiae show periosteal resorption. Which of the following serum components is most likely responsible for this patient's symptoms?

A) Calcitonin
B) Epinephrine
C) Parathyroid hormone
D) Prostacyclin
E) Type A natriuretic peptide

40. Five days after undergoing surgical repair of a middle cerebral artery aneurysm, a 52-year-old man has the onset of progressive lethargy. During his initial postoperative course, he was alert and only had mild right hemiparesis. Currently, he is oriented to person but not to place or time. His temperature is 37.8°C (100°F), pulse is 75/min, respirations are 20/min, and blood pressure is 130/80 mm Hg. Neurologic examination shows no new focal findings. Laboratory studies show:

| | |
|---|---|
| Leukocyte count | $7500/mm^3$ with a normal differential |
| Platelet count | $250,000/mm^3$ |
| Serum | |
| $Na^+$ | 115 mEq/L |
| $Cl^-$ | 90 mEq/L |
| $K^+$ | 3.5 mEq/L |
| Urea nitrogen | 20 mg/dL |
| Glucose, nonfasting | 130 mg/dL |
| Creatinine | 0.9 mg/dL |

A CT scan of the head shows no changes. Which of the following is the most appropriate next step in diagnosis?

A) Measurement of fasting serum glucose concentration
B) Measurement of serum cortisol concentration
C) Measurement of urine electrolyte concentrations
D) Blood cultures
E) Intracranial pressure monitoring

41. A 32-year-old woman comes to the physician because of a 3-week history of diffuse joint aches. Over the past 3 months, she has had headache, stiff neck, and progressive numbness, tingling, and weakness of both lower extremities. Neurologic examination shows a mild right facial palsy and mild weakness of the distal lower extremities. Sensation to pinprick and light touch is decreased over the lower extremities; deep tendon reflexes are absent. Which of the following is the most appropriate next step in diagnosis?

A) Measurement of serum Lyme (*Borrelia burgdorferi*) antibody concentration
B) Measurement of serum vitamin $B_{12}$ (cobalamin) concentration
C) Monospot test
D) PPD skin test
E) Rapid plasma reagin

42. A 67-year-old woman is brought to the emergency department 2 hours after the sudden loss of vision in her right eye. She has had severe intermittent headaches over the past week. She has a 3-month history of progressive fatigue and muscle aches that are most severe in her neck and shoulders. During this period, she has had temperatures to 38.3°C (100.9°F) and a 4.5-kg (10-lb) weight loss. She has a 10-year history of hypertension treated with lisinopril and osteoarthritis of the hips and knees treated with acetaminophen as needed. Her temperature is 38°C (100.4°F), pulse is 72/min and regular, respirations are 14/min, and blood pressure is 132/84 mm Hg. Funduscopic examination of the right eye shows pallor and edema of the optic disc; there is complete loss of vision. The temporal area and shoulders are tender. The remainder of the examination shows no abnormalities. Laboratory studies show:

| | |
|---|---|
| Hematocrit | 32% |
| Leukocyte count | 12,000/mm$^3$ with a normal differential |
| Erythrocyte sedimentation rate | 110 mm/h |
| Serum | |
| $Na^+$ | 136 mEq/L |
| $Cl^-$ | 104 mEq/L |
| $K^+$ | 4.2 mEq/L |
| $HCO_3^-$ | 24 mEq/L |
| Creatinine | 0.9 mg/dL |

Which of the following is the most likely underlying mechanism of this patient's symptoms?

A) Arterial atherosclerosis
B) Arterial embolization
C) Arterial inflammation
D) Arteriovenous malformation
E) Nerve degeneration
F) Nerve demyelination
G) Nerve infiltration
H) Venous embolization

43. A 78-year-old woman is admitted to the hospital because of a 3-day history of dizziness and weakness. She was diagnosed with chronic lymphocytic leukemia 2 years ago. Since then, she has had a 13.6-kg (30-lb) weight loss. She lives independently. She does not smoke cigarettes or drink alcohol. On admission, she appears confused and somewhat anxious. She is oriented to person but not to place or time. Her temperature is 36.4°C (97.5°F), pulse is 90/min, respirations are 12/min, and blood pressure is 110/60 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 94%. Physical examination shows lateral nystagmus and pallor. No rashes are noted. Cardiac examination shows a grade 2/6 systolic murmur heard best over the lower left sternal border. Romberg test is abnormal. Finger-nose testing is negative. Muscle strength is 4/5 throughout. Deep tendon reflexes are normal. She has an ataxic gait. Supplementation with which of the following is most likely to have prevented this patient's condition?

A) Folic acid
B) Vitamin $B_1$ (thiamine)
C) Vitamin $B_{12}$ (cyanocobalamin)
D) Vitamin C
E) Vitamin D

44. A 38-year-old man with nephrotic syndrome due to minimal change disease becomes weak and oliguric over a 4-day period. Treatment includes sodium restriction, prednisone, and furosemide. His blood pressure is 110/70 mm Hg while supine and 90/60 mm Hg while sitting. Examination shows no signs of systemic illness and no edema. Urinalysis shows 4+ protein, no blood, and 5–10 WBC/hpf. Which of the following is the most likely cause of the patient's weakness and oliguria?

A) Acute interstitial nephritis
B) Acute ischemic renal failure
C) Acute renal vein thrombosis
D) Intravascular volume depletion
E) Worsening of the nephrotic syndrome

45. A 23-year-old man with type 1 diabetes mellitus is admitted to the hospital because of a 2-day history of nausea, vomiting, and diarrhea. He also has had increased thirst and urination. During the past 2 hours, he has become tired and wants to sleep. He has not taken insulin in 24 hours. His temperature is 37.5°C (99.5°F), pulse is 102/min, respirations are 26/min, and blood pressure is 100/60 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 100%. Skin examination shows no abnormalities. The lungs are clear to auscultation. Heart sounds are normal. Bowel sounds are decreased; there is no organomegaly or tenderness. On neurologic examination, the patient is somnolent but easily aroused and fully oriented. Laboratory studies show:

| | |
|---|---|
| Hemoglobin | 15.5 g/dL |
| Hematocrit | 50% |
| Leukocyte count | 13,500/$mm^3$ |
| Segmented neutrophils | 70% |
| Lymphocytes | 30% |
| Serum | |
| $Na^+$ | 130 mEq/L |
| $K^+$ | 5.8 mEq/L |
| $Cl^-$ | 94 mEq/L |
| $HCO_3^-$ | 10 mEq/L |
| Urea nitrogen | 28 mg/dL |
| Glucose | 600 mg/dL |
| Creatinine | 1.2 mg/dL |

Arterial blood gas analysis on room air shows:

| | |
|---|---|
| pH | 7.13 |
| $Pco_2$ | 23 mm Hg |
| $Po_2$ | 98 mm Hg |

An x-ray of the chest shows no abnormalities. An ECG shows sinus tachycardia. This patient most likely has which of the following acid-base disorders?

A) Anion gap metabolic acidosis
B) Anion gap metabolic acidosis and metabolic alkalosis
C) Anion gap metabolic acidosis and respiratory acidosis

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| | |
|---|---|
| pH | 7.13 |
| $PCO_2$ | 23 mm Hg |
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A) Anion gap metabolic acidosis
B) Anion gap metabolic acidosis and metabolic alkalosis
C) Anion gap metabolic acidosis and respiratory acidosis
D) Respiratory acidosis and metabolic alkalosis
E) Respiratory alkalosis and metabolic acidosis

46. A 72-year-old woman comes to the emergency department because of an 8-hour history of severe, "crushing," left-sided chest pain that radiates to her neck and arms. She has no history of similar chest pain. During this time, she also has had shortness of breath and dizziness. She has hypertension and hyperlipidemia. She has no known allergies. Her medications are hydrochlorothiazide and lovastatin. She appears uncomfortable and in moderate distress. Her pulse is 120/min, and blood pressure is 98/60 mm Hg. Pulse oximetry on 6 L/min of oxygen by nasal cannula shows an oxygen saturation of 92%. Jugular venous pressure is 10 cm $H_2O$ (N=5–9). Carotid pulses are 2+ and equal with normal upstrokes. Crackles are heard halfway up the lungs bilaterally. Cardiac examination shows an $S_3$ gallop. The abdomen is nontender. There is 1+ edema to the mid calf bilaterally. Which of the following is the most likely diagnosis?

A) Acute aortic insufficiency
B) Acute papillary muscle rupture
C) Cardiac tamponade
D) Cardiogenic shock
E) Hypovolemia

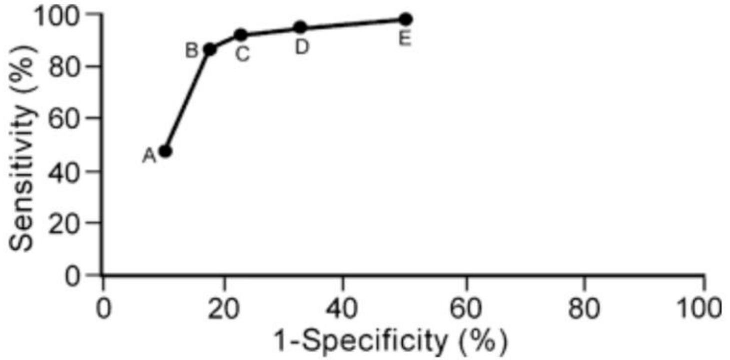
47. A 67-year-old man comes to the physician 6 hours after the onset of fever and lethargy. He has had dysuria for 3 days. He takes no medications. He is somnolent. His temperature is 38.9°C (102°F), pulse is 106/min, respirations are 36/min, and blood pressure is 88/60 mm Hg. The mucous membranes are dry, and the neck is supple. The lungs are clear to auscultation and percussion. Laboratory studies show:

| | |
|---|---|
| Hematocrit | 32% |
| Leukocyte count | 4800/$mm^3$ |
| Segmented neutrophils | 67% |
| Bands | 22% |
| Platelet count | 70,000/$mm^3$ |

Urinalysis shows numerous leukocytes and gram-negative bacilli. An x-ray of the chest shows no abnormalities. Blood cultures are ordered. In addition to intravenous administration of 0.9% saline, which of the following should be administered intravenously?

A) Antibiotics
B) Endotoxin antibody
C) Methylprednisolone
D) Platelets
E) Whole blood

48. A study is conducted to assess the usefulness of serum brain natriuretic peptide (BNP) concentrations as a diagnostic test for congestive heart failure (CHF). Subjects include patients older than 18 years of age who come to the emergency department because of shortness of breath. A serum BNP concentration is measured in each patient, and a clinical diagnosis of CHF is determined by two cardiologists through review of patients' medical records, chest x-rays on admission, and results of bedside echocardiography. The graph shows the sensitivity and specificity of BNP for determining the presence of CHF at five different serum BNP concentrations (represented by points A to E). At which labeled point does this test perform with the best combination of sensitivity and specificity?




A)
B)
C)
D)
E)

49. A 42-year-old woman comes to the office for a routine health maintenance examination. She has no history of serious illness and takes no medications. There is no family history of cancer. Physical examination, including pelvic examination, shows no abnormalities. The patient asks the physician about testing for ovarian cancer. Her friend was recently diagnosed with ovarian cancer, and the patient heard about a blood test to detect it. Research on this test shows a likelihood ratio of 1.0. Which of the following is the most appropriate physician response?

A) "If the test results are negative, you do not have cancer. A positive result is inconclusive."
B) "If the test results are positive, you have cancer. If they are negative, you do not have cancer."
C) "If the test results are positive, you have cancer. A negative result is inconclusive."
D) "The test can tell us if further testing is indicated for ovarian cancer."
E) "This test does not appear to be useful for diagnosing ovarian cancer."

50. A 77-year-old woman with a 2-year history of stage IV ovarian cancer is brought to the physician by her children for consultation. Initially, she underwent cytoreduction and combination chemotherapy with carboplatin and paclitaxel. Six months after completing chemotherapy, testing showed that the cancer had recurred. She now has metastatic disease to the spine despite three different chemotherapy regimens (doxorubicin, topotecan, and cyclophosphamide) over the past year. She has been confined to bed. She appears extremely ill and is in respiratory distress. She is alert and oriented and rates her pain as a 10 on a 10-point scale. Examination shows large supraclavicular lymph nodes and bilateral pleural effusions. Abdominal examination shows distention with ascites and a large, fixed ventral mass. She does not want further cancer therapy, but her family demands further treatment. Which of the following is the most appropriate next step?

A) Consult with the hospital ethics committee
B) Obtain psychiatric assessment
C) Intravenous morphine therapy
D) Intravenous chemotherapy
E) Operative cytoreduction